Marco Rando

POST-DOCTORAL RESEARCHER · MATHEMATICS & COMPUTER SCIENCE

Université Côte d'Azur, Inria, CNRS, LJAD, Nice, France

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Education

INRIA & CNRS-LJAD, Université Côte d'Azur *Nice*
POSTDOC IN MATHEMATICS AND COMPUTER SCIENCE *01/11/2025 - 31/07/2027*
- Research Field: Zeroth-order Bilevel Optimization
- Advisors: Prof. Samuel Vaiter, Prof. Jean-Baptiste Caillau

Malga - DIBRIS, University of Genoa *Genoa*
POSTDOC IN COMPUTER SCIENCE *01/11/2024 - 01/11/2025*
- Research Field: Black-box Optimization
- Advisor: Prof. Lorenzo Rosasco

Malga - DIBRIS, University of Genoa *Genoa*
PHD IN COMPUTER SCIENCE *01/11/2020 - 31/05/2024*
- Research Field: Black-box Optimization
- Final Evaluation: Excellent cum laude
- Advisor: Prof. Lorenzo Rosasco

DIBRIS, University of Genoa *Genoa*
MS DEGREE IN COMPUTER SCIENCE *01/11/2018 - 30/07/2020*
- Thesis Title: Bayesian Optimization for Parameter Tuning
- Final Evaluation: 110/110 cum laude
- Thesis Advisor: Prof. Lorenzo Rosasco

DIBRIS, University of Genoa *Genoa*
BS DEGREE IN COMPUTER SCIENCE *01/11/2015 - 26/07/2018*
- Thesis Title: The Consensus Problem in Distributed Systems: from theory to practice.
- Final Evaluation: 110/110 cum laude
- Thesis Advisor: Prof. Giorgio Delzanno

Research Areas & Current Research

MAIN RESEARCH AREAS

Zeroth-order Optimization. Development, theoretical analysis, and computational implementation of optimization methods for settings where only function evaluations, often noisy, are available. Research interests include stochastic and deterministic methods for convex and non-convex, smooth and non-smooth optimization, with particular emphasis on finite-difference methods, trust-region methods, Bayesian optimization, and metaheuristic techniques.

Langevin Methods for Non-convex Optimization. Development, theoretical analysis, and computational implementation of Langevin-based optimization methods for non-convex problems, including zeroth-order, stochastic, and non-smooth settings. Particular interests include convergence and complexity analysis, algorithm design, and the use of Langevin dynamics for efficient exploration of non-convex landscapes.

Optimization for Large Language Models. Development, theoretical analysis, and computational implementation of efficient optimization methods for large language models and other generative models. Research interests include zeroth- and first-order methods for fine-tuning and training, efficient computation and estimation of stochastic gradients for autoregressive probabilistic models, and optimization-based adversarial attacks and robustness of large language models.

Reinforcement Learning for Navigation. Investigation, implementation, and empirical evaluation of reinforcement learning algorithms for solving navigation tasks, especially under partial observability. This includes designing, analyzing and implementing learning-based strategies that enable agents to operate effectively in environments with limited or indirect feedback.

Recent & Current Research

Efficient Zeroth-order Stochastic Bilevel Optimization. Development of efficient zeroth-order methods for stochastic bilevel optimization. The project focuses on developing and analyzing single-loop methods with efficient hypergradient estimates. These estimates are obtained through the use of delayed information and reuse of function evaluations to compute multiple quantities involved in the hypergradient, thereby reducing the per-iteration computational cost. Convergence and complexity of the proposed methods are analyzed, with optimal complexity achieved for a Hessian-free variant. Empirical validation on synthetic and real-world problems demonstrates improved computational performance over state-of-the-art methods. As a by-product, a novel subspace-based adversarial attack is developed as an application of the proposed techniques.

Zeroth-order Langevin Algorithms for Non-convex Optimization. This project investigates Langevin-based zeroth-order methods for non-convex optimization, with the goal of deriving convergence rates under several settings. A particular focus is placed on deriving improved convergence rates for Langevin algorithms in both first- and zeroth-order settings. The work also introduces and analyzes the first structured zeroth-order Langevin method, with particular emphasis on its convergence properties and applicability to non-smooth optimization.

Fine-tuning and Jailbreaking Large Language Models. Development and analysis of efficient optimization methods for fine-tuning and adversarial prompting of large language models. For fine-tuning, particular focus is placed on zeroth-order Natural Evolution Strategies for settings with SFT-like target information, aiming to develop memory-efficient methods, and on fast first-order methods for RLHF to reduce the computational cost of sampling. For jailbreaking, optimization-based methods are investigated for tuning auxiliary models to generate sequences that induce target models to exhibit adversarial behavior, including first-order approaches inspired by RLHF. The project also introduces and studies the *Junking* regime, where non-semantic token sequences are optimized to induce target behaviors without relying on semantic instructions or prompt structure.

Efficient and Scalable Large-scale Kernel Methods. Development of scalable kernel methods for large-scale machine learning by addressing the memory bottleneck of preconditioned iterative methods for kernel ridge regression. The project introduces a novel, lightweight, and highly parallelizable preconditioner with memory requirements independent of both data and model size, and combines it with conjugate gradient to develop a new approach that enables kernel methods to scale to large datasets and models.

Variance Reduction for Structured Zeroth-order Methods. Development and analysis of variance-reduced finite-difference methods with structured sampling directions for non-smooth finite-sum optimization. The project investigates how structure in the directions, such as orthogonality, can be combined with variance-reduction techniques to improve the practical and theoretical performance of zeroth-order methods.

Ablation Study on Structured Finite-difference Methods. This research project explores the role of structured directional choices in finite-difference gradient estimation. We conduct an ablation study to evaluate how different sets of directions affect both the approximation quality and the convergence speed of the method. The goal is to gain deeper insight into the influence of structure in zeroth-order optimization techniques.

Q-Learning for Olfactory Navigation. In this research project, we investigate a reinforcement learning framework for olfactory navigation, where an agent must locate an odor source without access to its own spatial coordinates or the source's position. The only feedback available is the odor intensity at the agent's current location. We propose and analyze a Q-learning based approach tailored to this challenging partial-information setting.

Selected Publications

Rando M., Demetrio L., Rosasco L. & Roli F. **A new formulation for zeroth-order optimization of adversarial EXEmples in malware detection**. IEEE Transactions on Information Forensics and Security, 21, 506-515. (2025).

Rando M., Molinari C., Rosasco L. & Villa S. **An Optimal Structured Zeroth-order Algorithm for Non-smooth Optimization.** Advances in Neural Information Processing Systems, 36.

Rando M., Molinari C., Villa S. & Rosasco L. **Stochastic zeroth order descent with structured directions**. Computational Optimization and Applications, 1-37.

Rando M., Carratino C., Villa S. & Rosasco L. **Ada-BKB: Scalable Gaussian Process Optimization on Continuous Domains by Adaptive Discretization.** Proceedings of The 25th International Conference on Artificial Intelligence and Statistics, PMLR 151:7320-7348, (2022).

Rando M., James M., Verri A., Rosasco L. & Seminara A. **Q-learning with temporal memory to navigate turbulence**. eLife Sciences Publications, Ltd. 13:RP102906.

Sartore C., Rando M., Romualdi G., Molinari C., Rosasco L. & Pucci D. **Automatic gain tuning for humanoid robots walking architectures using gradient-free optimization techniques**. In 2024 IEEE-RAS 23rd International Conference on Humanoid Robots (Humanoids) (pp. 996-1003). IEEE.

Alfano P. D., Rando M., Letizia M., Odone F., Rosasco L. & Pastore V.P. **Efficient Unsupervised Learning for Plankton Images**. 26th International Conference on Pattern Recognition (ICPR), 1314-1321 (2022).

Letizia M., Losapio G., Rando M., Grosso G., Wulzer A., Pierini M., Zanetti M. & Rosasco L. **Learning new physics efficiently with nonparametric methods.** Eur. Phys. J. C 82, 879 (2022).

Grosso G., Lai N., Letizia M., Pazzini J., Rando M., Rosasco L., Wulzer A. & Zanetti M. **Fast kernel methods for data quality monitoring as a goodness-of-fit test.** Mach. Learn.: Sci. Technol. 4 035029 (2023).

Papers under Review

Rando M., & Vaiter S. **On the Hardness of Junking LLMs**. Under review in NeurIPS 2026.

Rando M., & Vaiter S. **ZOBA: An Efficient Single-loop Zeroth-order Bilevel Optimization Algorithm**. Under review in NeurIPS 2026.

Rando M., Heinonen R. A., Qi Y., & Seminara A. **Clock-state olfactory search in turbulent flows using Q-learning: The geometry of plume recovery**. Under review in Journal of the Royal Society Interface.

Piro L., Carbone M., Biferale L., Cencini M., Heinonen R. A., Rando M., & Seminara A. **Smart strategies to navigate turbulent odor plumes reorienting to local wind**. Under review in Physical Review E.

Rando M., Molinari C., Rosasco L. & Villa S. **A Structured Tour of Optimization with Finite Differences**. Under review in Mathematical Programming Computation.

Rando M., Traoré C., Molinari C., Rosasco L. & Villa S. **A Structured Proximal Stochastic Variance Reduced Zeroth-order Algorithm**. Under review in Computational Optimization and Applications.

Workshop Proceedings

Letizia M., Losapio G., Rando M., Grosso G. & Rosasco L. **Efficient kernel methods for model-independent new physics searches.** Proceedings of the deep learning for physical sciences workshop at NeurIPS (2021).

Teaching Experience

Topics in Modern Machine Learning *DIBRIS, University of Genoa*
POSITION: TEACHING ASSISTANT *19 June 2023 - 23 June 2023*

- Instructors: Lorenzo Rosasco, Nicoletta Noceti, Alessandro Verri, Simone Di Marino, Silvia Villa, Agnese Seminara,Cesare Molinari, Antoine Chatalic, Nicholas Schreuder

Ottimizzazione e Ricerca Operativa *DIMA, University of Genoa*
POSITION: TEACHING ASSISTANT *Feb 2023 - May 2023*

- Instructors: Prof. Silvia Villa

Ottimizzazione e Ricerca Operativa *DIMA, University of Genoa*
POSITION: TEACHING ASSISTANT *Feb 2024 - May 2024*

- Instructors: Prof. Silvia Villa

Ottimizzazione e Ricerca Operativa *DIMA, University of Genoa*
POSITION: TEACHING ASSISTANT *Feb 2025 - May 2025*

- Instructors: Prof. Silvia Villa

Machine Learning Crash Course *DIBRIS, University of Genoa*
POSITION: TEACHING ASSISTANT *27 June 2022 - 01 July 2022*

- Instructors: Prof. Lorenzo Rosasco, Prof. Giovanni S. Alberti, Prof. Simone Di Marino

A Tour of Reinforcement Learning and Applications *DICCA, University of Genoa*
POSITION: TEACHING ASSISTANT *30 May 2022 - 01 July 2022*

- Instructors: Prof. Agnese Seminara

Advanced Machine Learning *DIBRIS, University of Genoa*
POSITION: TEACHING ASSISTANT *Feb 2023 - May 2023*

- Instructors: Prof. Lorenzo Rosasco, Prof. Nicoletta Noceti, Prof. Alessandro Verri

Advanced Machine Learning *DIBRIS, University of Genoa*
POSITION: TEACHING ASSISTANT *Mar 2022 - May 2022*

- Instructors: Prof. Lorenzo Rosasco, Prof. Nicoletta Noceti, Prof. Alessandro Verri

Machine Learning *DIBRIS, University of Genoa*
POSITION: TEACHING ASSISTANT *Oct 2021 - Dec 2021*

- Instructors: Prof. Lorenzo Rosasco, Prof. Nicoletta Noceti

Machine Learning *DIBRIS, University of Genoa*
POSITION: TEACHING ASSISTANT *Sep 2022 - Dec 2022*

- Instructors: Prof. Lorenzo Rosasco, Prof. Nicoletta Noceti

Event & Workshop

EVENTS/WORKSHOP/SCHOOLS ORGANIZATION

Theoretical Foundations of Machine Learning Course *Genoa, Italy*
POSITION: ORGANIZER *23/06/2025 - 27/06/2025*

WORKSHOP PARTECIPATIONS

Ellis Workshop "UnWorkshop on Algorithms for Deep Learning" *Tubingen, Germany*
18/07/2023 - 19/07/2023

CSW23: Unige 4th PhD Computer Science Workshop *Genoa, Italy*
31/05/2023 - 31/05/2023

Ellis Theory Workshop *Arenzano, Italy*
20/06/2022 - 22/06/2022

CSW22: Unige 3rd PhD Computer Science Workshop *Genoa, Italy*
09/06/2022 - 09/06/2022

Invited Talks and Presentations

PEPR IA Days 2026 *Rennes, FR*
Presentation: Towards Efficient Zeroth-order Bilevel Optimization *22/06/2026 - 24/06/2026*

The 5th Italian Meeting on Probability and Mathematical Statistics 2026 *Palermo, IT*
Presentation: ZOBA: An Efficient Single-loop Zeroth-order Bilevel Optimization Algorithm *8/06/2026 - 12/06/2026*

SMAI-MODE 2026 *Nice, FR*
Presentation: Efficient Zeroth-order Bilevel Optimization *18/03/2026 - 20/03/2026*

EURO 2025 *Leeds, UK*
Presentation: Structured Finite-Difference Methods for Zeroth-order Optimization *22/06/2025 - 25/06/2025*

EUROPT 2024 *Lund, SE*
Presentation: Zeroth order descent with Structured Directions *26/06/2024 - 28/06/2024*

NP-Twins 2024 *Genoa, IT*
Presentation: Recent developments in zero-th order optimization algorithms *16/12/2024 - 18/12/2024*

Peer Review

INFORMS: Journal on Computing.

Pattern Recognit.: Pattern Recognition.

FnT Optim: Foundations and Trends in Optimization.

AAMAS 2025: The 24th International Conference on Autonomous Agents and Multi-Agent Systems.

NeurIPS (2023/2024/2025/2026): The Annual Conference on Neural Information Processing Systems.

AISTATS (2022/2023/2024/2025) The International Conference on Artificial Intelligence and Statistics.

ICML 2026: International Conference on Machine Learning.

JMLR: Journal of Machine Learning Research.

IMAJNA: Journal of Numerical Analysis.

Awards, Fellowships, & Grants

2026 **ICML Silver Reviewer Award**, ICML 2026
2023 **AISTATS Top Reviewer Award**, AISTATS 2023

Mentoring

Giordano Vitale, Master Student in Mathematics, Università degli Studi di Milano Bicocca
Amir Elarbi, Intership at LJAD, CNRS, Université Côte d’Azur